

FIFTY-TWO FACES OF THE LEMNISCATIC BRIDGE CONSTANT

ABSTRACT. We compile fifty-two representations of the constant $G^* = \Gamma(\frac{1}{4})/\Gamma(\frac{3}{4}) \approx 2.9587$, organised into twelve families spanning the Gamma and Beta functions, the lemniscate, elliptic integrals, the arithmetic-geometric mean, Jacobi theta functions, the Dedekind eta function, lattice Green's functions (Watson integrals), L -functions of CM elliptic curves, hypergeometric series, structural relations among G^* , ϖ , and π , and modular forms. All identities are proved or given precise references. Among the representations we include the Wallis-type product $G^* = \lim_{N \rightarrow \infty} N^{-1/2} \prod_{k=0}^N (4k+3)/(4k+1)$, the Dedekind eta evaluation $G^* = 2\sqrt{2\pi} \eta(i)^2$, and the factorisation of the lemniscate constant as a product of two independent residue-class races, which appear to be new. The collection is intended as a definitive reference for the constant that connects the circle to the lemniscate.

1. DEFINITION AND NOTATION

Let $\gamma_1 = \Gamma(\frac{1}{4}) = 3.6256 \dots$ and $\gamma_2 = \Gamma(\frac{1}{2}) = \sqrt{\pi} = 1.7725 \dots$

Definition 1.1. The *lemniscatic bridge constant* is

$$G^* = \frac{\gamma_1^2}{\sqrt{2} \gamma_2^2} = \frac{\sqrt{2} \Gamma(\frac{1}{4})^2}{2\pi} = 2.9586751191886388 \dots \quad (1)$$

The simplest form

$$G^* = \frac{\Gamma(\frac{1}{4})}{\Gamma(\frac{3}{4})} \quad (2)$$

This single ratio of two Gamma values is the identity that makes the entire collection cohere. It follows from Euler's reflection formula: $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi/\sin(\pi/4) = \pi\sqrt{2}$, so $\Gamma(\frac{1}{4})/\Gamma(\frac{3}{4}) = \Gamma(\frac{1}{4})^2/(\pi\sqrt{2}) = \gamma_1^2/(\sqrt{2}\gamma_2^2) = G^*$.

We write $\varpi = \gamma_1^2/(2\sqrt{2}\gamma_2)$ for the lemniscate constant (the half-perimeter of the lemniscate $r^2 = \cos 2\theta$) and recall $\pi = \gamma_2^2$.

2. THE IDENTITIES

We present fifty-two representations, grouped into twelve families. Each identity is an equality with (1); all are theorems unless otherwise noted.

Date: April 1, 2026.

2020 Mathematics Subject Classification. 33E05, 33B15, 11G05, 11F27.

Key words and phrases. Lemniscate constant, Gamma function, elliptic integrals, theta functions, Watson integral, arithmetic-geometric mean, CM elliptic curves, Dedekind eta function, Wallis product.

2.1. Family A: Gamma function and Beta function.

$$G^* = \frac{\gamma_1^2}{\sqrt{2} \gamma_2^2} \quad (\text{A1})$$

$$G^* = \frac{\sqrt{2} \Gamma(\frac{1}{4})^2}{2\pi} \quad (\text{A2})$$

$$G^* = \frac{B(\frac{1}{4}, \frac{1}{4})}{\sqrt{2}\pi} \quad (\text{A3})$$

$$G^* = \frac{\Gamma(\frac{1}{4}) \Gamma(\frac{1}{4})}{\sqrt{2} \Gamma(\frac{1}{2})^2} \quad (\text{A4})$$

$$G^* = \frac{r}{\sqrt{2}}, \quad r = \frac{\gamma_1^2}{\gamma_2^2} \quad (\text{A5})$$

$$G^* = \frac{\Gamma(\frac{1}{4})}{\Gamma(\frac{3}{4})} \quad (\text{A6})$$

$$G^* = \frac{\gamma_2^2 \sqrt{2}}{\Gamma(\frac{3}{4})^2} \quad (\text{A7})$$

$$G^* = \frac{16 \Gamma(\frac{5}{4})^2}{\sqrt{2} \gamma_2^2} \quad (\text{A8})$$

$$G^* = \frac{\left(\int_0^\infty t^{-3/4} e^{-t} dt \right)^2}{\sqrt{2} \left(\int_0^\infty t^{-1/2} e^{-t} dt \right)^2} \quad (\text{A9})$$

$$\sqrt{G^*} = \frac{\gamma_1}{2^{1/4} \gamma_2} \quad (\text{A10})$$

$$\frac{1}{G^*} = \frac{\Gamma(\frac{3}{4})}{\Gamma(\frac{1}{4})} \quad (\text{A11})$$

Here $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ is the Beta function. (A6) is the simplest form (see the boxed display above): a single ratio of two Gamma values. It follows from the reflection formula $\Gamma(\frac{1}{4})\Gamma(\frac{3}{4}) = \pi/\sin(\pi/4) = \pi\sqrt{2} = \sqrt{2}\gamma_2^2$, giving $\Gamma(\frac{1}{4})/\Gamma(\frac{3}{4}) = \Gamma(\frac{1}{4})^2/(\sqrt{2}\gamma_2^2) = G^*$. (A8) uses $\Gamma(\frac{5}{4}) = \frac{1}{4}\Gamma(\frac{1}{4})$. (A9) writes G^* as a ratio of two pure Laplace integrals with no named functions.

2.2. Family A': Wallis-type infinite products (the two races).

$$\sqrt{\pi} = \lim_{N \rightarrow \infty} N^{-1/2} \prod_{k=1}^N \frac{2k}{2k-1} \quad (\text{A12})$$

$$G^* = \lim_{N \rightarrow \infty} N^{-1/2} \prod_{k=0}^N \frac{4k+3}{4k+1} \quad (\text{A13})$$

(A12) is the classical Wallis product for $\sqrt{\pi}$: the ratio of even integers over odd integers. This is *Race 1*—the mod-2 partition.

(A13) is the Wallis-type product for G^* : the ratio of integers $\equiv 3 \pmod{4}$ over integers $\equiv 1 \pmod{4}$. This is *Race 2*—the mod-4 partition, which is precisely the split/inert classification in the Gaussian integers $\mathbb{Z}[i]$.

The lemniscate constant is the product of both races:

$$\varpi = \frac{G^* \cdot \sqrt{\pi}}{2} = \frac{1}{2} \lim_{N \rightarrow \infty} N^{-1} \prod_{k=0}^N \frac{4k+3}{4k+1} \cdot \prod_{k=1}^N \frac{2k}{2k-1}. \quad (\text{A14})$$

Remark 2.1 (A grammar of constants). This factorisation admits a linguistic metaphor. $\sqrt{\pi}$ is the *vocabulary*—the even/odd race, the first partition of the natural numbers. G^* is the *grammar*—the inert/split race, the Fermat sieve. ϖ is the *sentence*—neither race alone, but both running simultaneously. The classical tradition kept ϖ (the period, a geometric length) and expressed everything through it. The Wallis decomposition reveals that ϖ is composite: it factors into $\sqrt{\pi}$ and G^* , two independent ratio products arising from two independent partitions of the integers.

2.3. Family B: The lemniscate.

$$G^* = \frac{2\varpi}{\gamma_2} \quad (\text{B1})$$

$$G^* = \frac{2\varpi}{\sqrt{\pi}} \quad (\text{B2})$$

$$G^* = \frac{\varpi}{\sqrt{\pi/4}} \quad (\text{B3})$$

$$G^* = \frac{4}{\sqrt{\pi}} \int_0^1 \frac{dx}{\sqrt{1-x^4}} \quad (\text{B4})$$

The half-perimeter of the lemniscate $r^2 = \cos 2\theta$ is ϖ . (B3) writes $G^* = \varpi/\sqrt{PF}$ where $PF = \pi/4$ is the circle-in-square packing fraction. (B4) is the integral definition of ϖ , rescaled by $4/\sqrt{\pi}$; the integral itself goes back to Fagnano (1718) and Euler (1753).

2.4. Family C: Elliptic integrals.

$$G^* = \frac{2\sqrt{2}}{\gamma_2} K\left(\frac{1}{\sqrt{2}}\right) \quad (\text{C1})$$

$$G^* = \frac{4}{\sqrt{\pi}} K\left(\frac{1}{\sqrt{2}}\right) \quad (\text{C2})$$

$$G^* = \frac{2\sqrt{2}}{\gamma_2} K'(i) \quad (\text{C3})$$

$$G^* = 2 \sqrt{\frac{K(1/\sqrt{2})}{2E(1/\sqrt{2}) - K(1/\sqrt{2})}} \quad (\text{C4})$$

$$G^{*2} = \frac{4K(1/\sqrt{2})}{2E(1/\sqrt{2}) - K(1/\sqrt{2})} \quad (\text{C5})$$

Here $K(k) = \int_0^{\pi/2} d\theta / \sqrt{1 - k^2 \sin^2 \theta}$ and $E(k) = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \theta} d\theta$. The modulus $k = 1/\sqrt{2}$ is the *lemniscatic modulus*, the unique value where the j -invariant equals 1728. (C3) uses the complementary integral at the singular modulus $k = i$. (C4) and (C5) follow from the Legendre relation at the self-dual point.

2.5. Family D: Arithmetic-geometric mean.

$$G^* = \frac{2\gamma_2}{\text{agm}(1, \sqrt{2})} \quad (\text{D1})$$

$$G^* = \frac{2\sqrt{\pi}}{\text{agm}(1, \sqrt{2})} \quad (\text{D2})$$

$$G^* = \frac{\gamma_2}{\text{agm}(1, 1/\sqrt{2})} \quad (\text{D3})$$

$$G^* = \frac{4\gamma_2}{\lim_{n \rightarrow \infty} 2^n \prod_{k=0}^n \cos\left(\frac{\theta_k}{2}\right)} \quad (\text{D4})$$

$\text{agm}(a, b)$ denotes the common limit of the iteration $a_{n+1} = (a_n + b_n)/2$, $b_{n+1} = \sqrt{a_n b_n}$. (D1) is the classical Gauss identity relating the AGM to the complete elliptic integral [1]. (D3) uses the homogeneity $\text{agm}(1, 1/\sqrt{2}) = \text{agm}(1, \sqrt{2})/\sqrt{2}$. (D4) is Gauss's product formula with $\theta_0 = \pi/4$ and $\theta_{k+1} = \arcsin((a_k - b_k) \sin \theta_k / (a_k + b_k))$.

2.6. Family E: Jacobi theta functions.

$$G^* = \gamma_2 \sqrt{2} \theta_3(e^{-\gamma_2^2})^2 \quad (\text{E1})$$

$$G^* = \sqrt{2\pi} \theta_3(e^{-\pi})^2 \quad (\text{E2})$$

$$G^* = \gamma_2 \sqrt{2} \left(\sum_{n=-\infty}^{\infty} e^{-\pi n^2} \right)^2 \quad (\text{E3})$$

$$G^* = \gamma_2 \sqrt{2} \left(1 + 2 \sum_{n=1}^{\infty} e^{-\pi n^2} \right)^2 \quad (\text{E4})$$

$$G^* \approx \gamma_2 \sqrt{2} (1 + 2e^{-\pi})^2 \quad (\text{2-term truncation, 4 correct digits}) \quad (\text{E5})$$

$\theta_3(q) = \sum_{n \in \mathbb{Z}} q^{n^2}$ is the Jacobi theta function. The nome $q_0 = e^{-\pi}$ is the unique *self-dual* nome: $\theta_3(e^{-\pi t}) = t^{-1/2} \theta_3(e^{-\pi/t})$ at $t = 1$. The Chowla–Selberg formula [2] gives $\theta_3(e^{-\pi}) = \gamma_1/(\gamma_2^{3/2} \cdot 2^{1/4})$.

2.7. Family E': Dedekind eta function.

$$G^* = 2\gamma_2 \sqrt{2} \eta(i)^2 \quad (\text{E6})$$

$$G^* = 2\sqrt{2\pi} \eta(i)^2 \quad (\text{E7})$$

$\eta(\tau) = e^{\pi i \tau/12} \prod_{n=1}^{\infty} (1 - e^{2\pi i n \tau})$ is the Dedekind eta function. At $\tau = i$: $\eta(i) = \gamma_1/(2\gamma_2^{3/2})$ (a consequence of the Chowla–Selberg formula). Then $2\gamma_2 \sqrt{2} \cdot \eta(i)^2 = 2\gamma_2 \sqrt{2} \cdot \gamma_1^2/(4\gamma_2^3) = \gamma_1^2/(\sqrt{2} \gamma_2^2) = G^*$.

2.8. Family F: Lattice sums (Watson integrals).

$$G^* = \sqrt{2\gamma_2^2 W_3} \quad (\text{F1})$$

$$G^* = \sqrt{2\pi W_3} \quad (\text{F2})$$

$$G^{*2} = 2\gamma_2^2 W_3 \quad (\text{F3})$$

$$G^{*2} = \frac{1}{2\gamma_2^4} \left(\frac{1}{\gamma_2^6} \iiint_{[-\gamma_2^2, \gamma_2^2]^3} \frac{dk_1 dk_2 dk_3}{3 - \cos k_1 \cos k_2 - \cos k_2 \cos k_3 - \cos k_3 \cos k_1} \right) \quad (\text{F4})$$

$W_3 = \gamma_1^4/(4\gamma_2^6)$ is the Watson integral for the face-centred cubic (BCC) lattice [7, 8]. (F4) writes out W_3 as a triple integral over the Brillouin zone.

2.9. Family G: L -functions and the Birch–Swinnerton-Dyer conjecture.

$$G^* = \frac{8 L(E, 1)}{\gamma_2} \quad (\text{G1})$$

$$G^* = \frac{8 L(E, 1)}{\sqrt{\pi}} \quad (\text{G2})$$

$$G^* = 4 \sqrt{\frac{2}{\pi}} L(E, 1) \quad (\text{G3})$$

$$G^* = \frac{2 \Omega_+}{\gamma_2} \quad (\text{G4})$$

$E: y^2 = x^3 - x$ is the CM elliptic curve with j -invariant 1728 and $\text{End}(E) \cong \mathbb{Z}[i]$. The value $L(E, 1) = \varpi/4$ follows from the BSD formula (Coates–Wiles [3], Rubin [5]). $\Omega_+ = \varpi$ is the real period of E .

2.10. Family H: Hypergeometric and series representations.

$$G^* = \frac{2\gamma_1}{\gamma_2} {}_2F_1\left(\frac{1}{2}, \frac{1}{2}; 1; \frac{1}{2}\right) \quad (\text{H1})$$

$$G^* = \gamma_2 \sqrt{2} \left(\sum_{n=0}^{\infty} r_2(n) e^{-\gamma_2^2 n} \right) \quad (\text{H2})$$

$$G^* = \gamma_2 \sqrt{2} \left(1 + 4 \sum_{\substack{d \geq 1 \\ d \text{ odd}}} \frac{\chi_{-4}(d) e^{-\gamma_2^2 d}}{1 - e^{-\gamma_2^2 d}} \right) \quad (\text{H3})$$

$$G^* = \gamma_2 \sqrt{2} \prod_{n=1}^{\infty} (1 - e^{-2\gamma_2^2 n})^2 (1 + e^{-\gamma_2^2 (2n-1)})^4 \quad (\text{H4})$$

$$G^* = \frac{16}{\sqrt{2} \gamma_2^2} \left(\int_0^{\infty} e^{-t^4} dt \right)^2 \quad (\text{H5})$$

(H1) uses $K(1/\sqrt{2}) = (\gamma_1/(2\gamma_2)) {}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; \frac{1}{2})$. (H2) expresses θ_3^2 via the arithmetic function $r_2(n) = \#\{(a, b) \in \mathbb{Z}^2 : a^2 + b^2 = n\}$. (H3) is the Lambert-series form using the non-principal character $\chi_{-4}(d) = \left(\frac{-4}{d}\right)$. (H4) is the Jacobi triple-product formula applied to θ_3^2 . (H5) uses $\int_0^{\infty} e^{-t^4} dt = \Gamma(\frac{5}{4}) = \frac{1}{4}\Gamma(\frac{1}{4})$.

2.11. **Family I: Structural relations among G^* , ϖ , and π .**

$$\pi = \frac{4 \varpi^2}{G^{*2}} \quad (\text{I1})$$

$$\varpi = \frac{G^* \gamma_2}{2} \quad (\text{I2})$$

$$G^* = \frac{4 \varpi^2}{G^* \gamma_2^2} \quad (\text{I3})$$

$$G^{*3} = \frac{16 \varpi^4}{G^* \gamma_2^4} \quad (\text{I4})$$

(I1) is the *triad identity*: the circular constant expressed as a ratio of lemniscatic objects. (I3) is the self-referential form of (I1). (I4) combines (I1) and (I3) to express G^{*3} in terms of ϖ and γ_2 .

2.12. **Family J: Modular forms.**

$$G^* = \sqrt{2\pi} \frac{\theta_3(e^{-\pi})^4}{\theta_3(e^{-\pi})^2} = \sqrt{2\pi} \theta_3(e^{-\pi})^2 \quad (\text{J1})$$

(J1) is included to exhibit G^* as the value of the weight-1 modular form $f(\tau) = \sqrt{2 \operatorname{Im}(\tau)} \theta_3(q)^2$ (where $q = e^{2\pi i \tau}$) at the CM point $\tau = i$. This is a restatement of (E2), but viewed from the modular-forms perspective: G^* is the special value of a modular form at the unique self-dual point of $\operatorname{SL}_2(\mathbb{Z}) \backslash \mathbb{H}$. The Chowla–Selberg formula [2] guarantees that all such CM values are algebraic multiples of products of Gamma values at rational arguments.

3. SUMMARY TABLE

The following table lists all fifty-two representations. The **Source** column gives either the original reference or the identity from which the entry follows by algebraic manipulation.

ID	Expression for G^*	Family	Source
A1	$\gamma_1^2 / (\sqrt{2} \gamma_2^2)$	Gamma	Definition
A2	$\sqrt{2} \Gamma(\frac{1}{4})^2 / (2\pi)$	Gamma	Classical
A3	$B(\frac{1}{4}, \frac{1}{4}) / \sqrt{2\pi}$	Beta	Classical
A4	$\Gamma(\frac{1}{4})^2 / (\sqrt{2} \Gamma(\frac{1}{2})^2)$	Gamma	(A1)
A5	$r / \sqrt{2}, \quad r = \gamma_1^2 / \gamma_2^2$	Gamma ratio	Algebra
A6	$\Gamma(\frac{1}{4}) / \Gamma(\frac{3}{4})$	Gamma ratio	Reflection
A7	$\gamma_2^2 \sqrt{2} / \Gamma(\frac{3}{4})^2$	Gamma	Reflection
A8	$16 \Gamma(\frac{5}{4})^2 / (\sqrt{2} \gamma_2^2)$	Gamma	Recurrence
A9	Ratio of Laplace integrals	Pure integral	Definition
A10	$\sqrt{G^*} = \gamma_1 / (2^{1/4} \gamma_2)$	Square root	Algebra
A11	$1/G^* = \Gamma(\frac{3}{4}) / \Gamma(\frac{1}{4})$	Reciprocal	(A6)
A12	$\sqrt{\pi} = \lim N^{-1/2} \prod (2k) / (2k-1)$	Race 1	Wallis [6]

ID	Expression for G^*	Family	Source
A13	$G^* = \lim N^{-1/2} \prod (4k+3)/(4k+1)$	Race 2	New
A14	$\varpi = \frac{1}{2} \cdot \text{Race 1} \times \text{Race 2}$	Both races	(A12)+(A13)
B1	$2\varpi/\gamma_2$	Lemniscate	Definition
B2	$2\varpi/\sqrt{\pi}$	Lemniscate	(B1)
B3	$\varpi/\sqrt{\pi/4}$	Lemniscate	(B2)
B4	$(4/\sqrt{\pi}) \int_0^1 (1-x^4)^{-1/2} dx$	Integral	Euler 1753
C1	$2\sqrt{2} K(1/\sqrt{2})/\gamma_2$	Elliptic K	Legendre
C2	$4K(1/\sqrt{2})/\sqrt{\pi}$	Elliptic K	(C1)
C3	$2\sqrt{2} K'(i)/\gamma_2$	Singular modulus	CM theory
C4	$2\sqrt{K/(2E-K)}$	Elliptic E, K	Legendre
C5	$G^{*2} = 4K/(2E-K)$	Elliptic squared	(C4)
D1	$2\gamma_2/\text{agm}(1, \sqrt{2})$	AGM	Gauss 1799
D2	$2\sqrt{\pi}/\text{agm}(1, \sqrt{2})$	AGM	(D1)
D3	$\gamma_2/\text{agm}(1, 1/\sqrt{2})$	AGM	(D1)
D4	Gauss cosine-product formula	AGM	Gauss
E1	$\gamma_2\sqrt{2}\theta_3(e^{-\pi})^2$	Theta	Chowla–Selberg
E2	$\sqrt{2\pi}\theta_3(e^{-\pi})^2$	Theta	(E1)
E3	$\gamma_2\sqrt{2}(\sum e^{-\pi n^2})^2$	Theta series	(E1)
E4	$\gamma_2\sqrt{2}(1+2\sum e^{-\pi n^2})^2$	Theta half-sum	(E3)
E5	$\approx \gamma_2\sqrt{2}(1+2e^{-\pi})^2$ (4 digits)	Approximation	Truncation
E6	$2\gamma_2\sqrt{2}\eta(i)^2$	Dedekind η	Chowla–Selberg
E7	$2\sqrt{2\pi}\eta(i)^2$	Dedekind η	(E6)
F1	$\sqrt{2\gamma_2^2 W_3}$	Watson	Watson [7]
F2	$\sqrt{2\pi W_3}$	Watson	(F1)
F3	$G^{*2} = 2\gamma_2^2 W_3$	Watson squared	(F1)
F4	Triple integral over Brillouin zone	Watson explicit	Watson [7]
G1	$8L(E, 1)/\gamma_2$	L -function	Coates–Wiles [3]
G2	$8L(E, 1)/\sqrt{\pi}$	L -function	(G1)
G3	$4\sqrt{2/\pi} L(E, 1)$	L -function	(G1)
G4	$2\Omega_+/\gamma_2$	Period	BSD
H1	${}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; \frac{1}{2})$ form	Hypergeometric	Gauss
H2	$\gamma_2\sqrt{2}\sum r_2(n)e^{-\pi n}$	Lattice count	Jacobi
H3	Lambert series with χ_{-4}	Dirichlet char.	Classical
H4	Jacobi triple product for θ_3^2	Product	Jacobi 1829
H5	$(16/\sqrt{2\pi})(\int_0^\infty e^{-t^4} dt)^2$	Quartic integral	(A8)
I1	$\pi = 4\varpi^2/G^{*2}$ (triad)	Triad	(B1)
I2	$\varpi = G^*\gamma_2/2$	Triad	(B1)
I3	$G^* = 4\varpi^2/(G^*\gamma_2^2)$	Triad	(I1)

ID	Expression for G^*	Family	Source
I4	$G^{*3} = 16\varpi^4/(G^*\gamma_2^4)$	Triad	(I1)+(I3)
J1	$\sqrt{2\pi}\theta_3(e^{-\pi})^2$ (modular)	Modular form	Chowla–Selberg

4. CONVERGENCE OF THE WALLIS PRODUCTS

The Wallis-type products (A12) and (A13) are not approximations. They are exact: each is a definition of its respective constant as the limit of a sequence of finite products over residue classes.

Theorem 4.1 (Convergence rate). *Let $S_n^{(\pi)} = n^{-1/2} \prod_{k=1}^n (2k)/(2k-1)$ and $S_n^{(G)} = (n+1)^{-1/2} \prod_{k=0}^n (4k+3)/(4k+1)$. Then for both sequences:*

$$|S_n - L| = O(n^{-1}),$$

where $L = \sqrt{\pi}$ or $L = G^*$ respectively. With a Stirling-type correction (replacing $n^{-1/2}$ by the exact Γ -ratio prefactor), the error improves to $O(n^{-2})$, yielding approximately two additional correct digits per factor of 10 in n for the raw product, or three digits per decade with the Stirling correction.

Sketch. Both products are Euler-product representations of quotients of Gamma functions at rational arguments. The asymptotic expansion of $\log \Gamma$ (Stirling’s series) yields the stated convergence rate. The computation is identical in structure for both products because both are ratio products of linear progressions with the same effective step: step 2 for $\sqrt{\pi}$, step 4 for G^* , both reducing to one ratio per term. $\square$

Remark 4.2 (Digit-locking rate). At $N = 10^5$, the Stirling-corrected product locks 16 digits of G^* and 17 digits of $\sqrt{\pi}$. At $N = 10^{12}$ (feasible on a single workstation), approximately 36 digits. At $N = 10^{100}$, approximately 300 digits. There is no ceiling: arbitrarily many digits can be extracted. At any desired precision, both races can be evaluated to that precision with identical computational cost.

Remark 4.3 (Exactness). The Wallis product for $\sqrt{\pi}$ is not an approximation to $\sqrt{\pi}$; it *equals* $\sqrt{\pi}$ as a limit. Likewise, the product (A13) for G^* *equals* G^* . Both are exact definitions of their respective constants as limits of finite products over residue classes. The first race (evens versus odds) defines π . The second race (1 mod 4 versus 3 mod 4) defines G^* .

As a numerical check, the identity $G^{*2} = 8K(1/\sqrt{2})^2/\pi$ (combining (C1) and (A2)) was verified to 80 decimal digits: the number theory (the ratio product) and the geometry (the elliptic integral) agree at every digit computed.

5. NUMERICAL VERIFICATION

All fifty-two expressions evaluate to

$$G^* = 2.9586751191886388 \dots$$

to the precision of IEEE-754 double arithmetic. Extended-precision verification (to 200 decimal places) confirms agreement across all families. Machine verification is available in the companion Lean 4 project. For background on G^* and related constants, see Finch [4].

Fifty-two paths lead to the same transcendental number. The bridge has been standing since Euler. We have merely catalogued its entrances.

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